

paper001

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Abstract

To be drafted.

1 Introduction

1. Introduction to gamma implications in the brain.
2. Summary of SOTA of modelling gamma.
3. We train Gamma for the first time.
4. It brings sparsity benefits by architecture.

2 Methods

1. Summary of single neuron equations.
2. Summary of network architecture.
3. Defense of fixed W_{ei} W_{ie} weights.
4. Summary of rhythmicity metric with literature review.
5. Summary of training.
6. Summary of gradient damping method for the PING case

3 Results

Paragraphs on recurrent loop creates Gamma. References Figure 1.

Paragraphs on supercritical Hopf bifurcation. References Figure 2.

Paragraphs on PING uses less E spikes. References Figure 3.

Paragraphs on sparsity is architecture enforced, not learned. References Figure 4 and 5.

Paragraphs on one spike per cell per cycle. References Figure 6, 7 and 8.

Paragraphs on the gating is dynamical and robust. References Figure 9, 10 and 11.

Paragraphs on real-time streaming classification. References Figure 12 and 13.

4 Discussion

To be drafted.

5 Conclusion

To be drafted.

A Figures

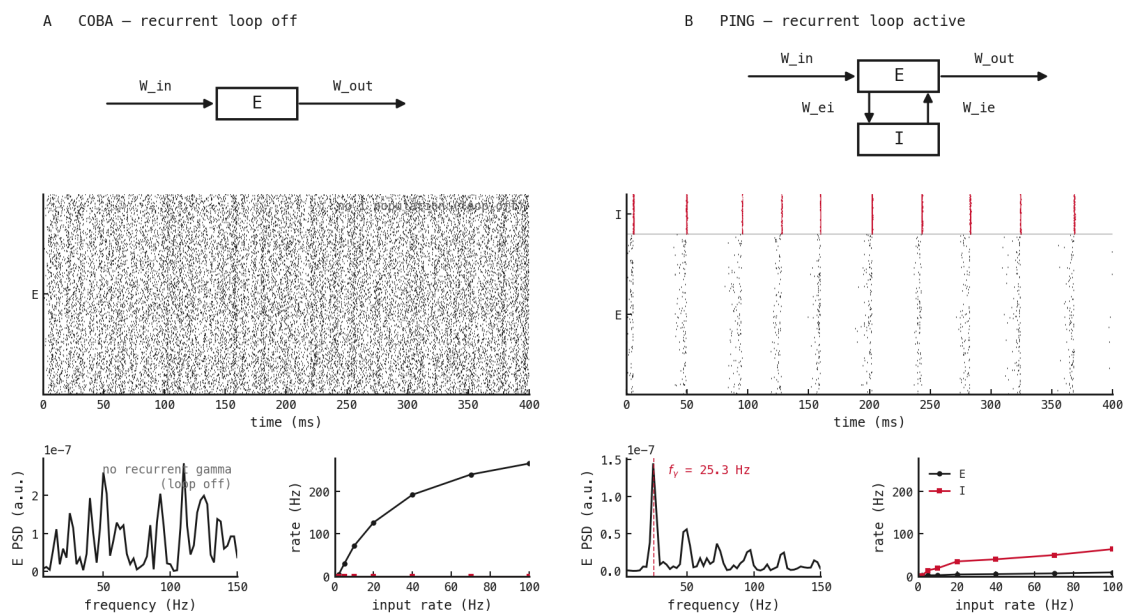


Figure 1:

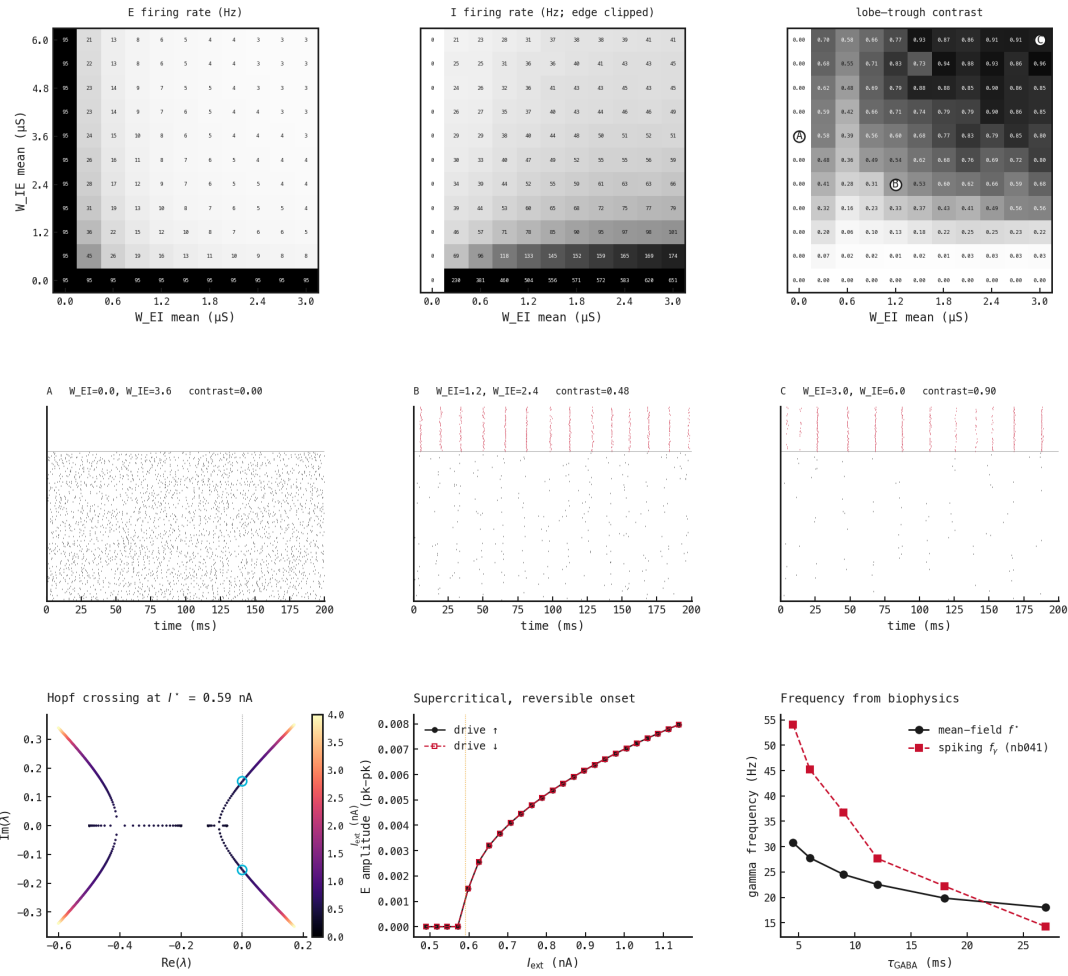


Figure 2:

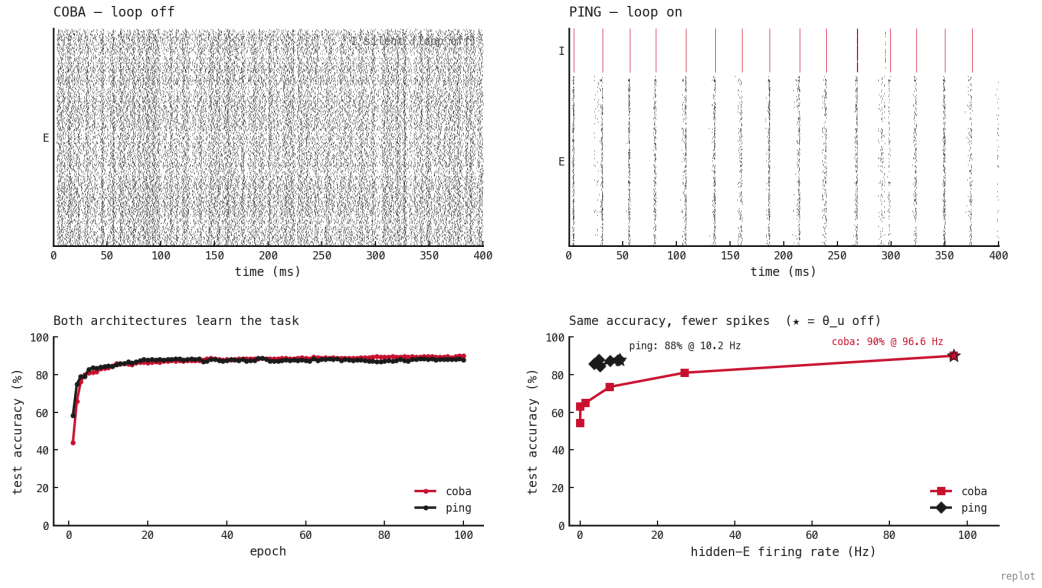


Figure 3:

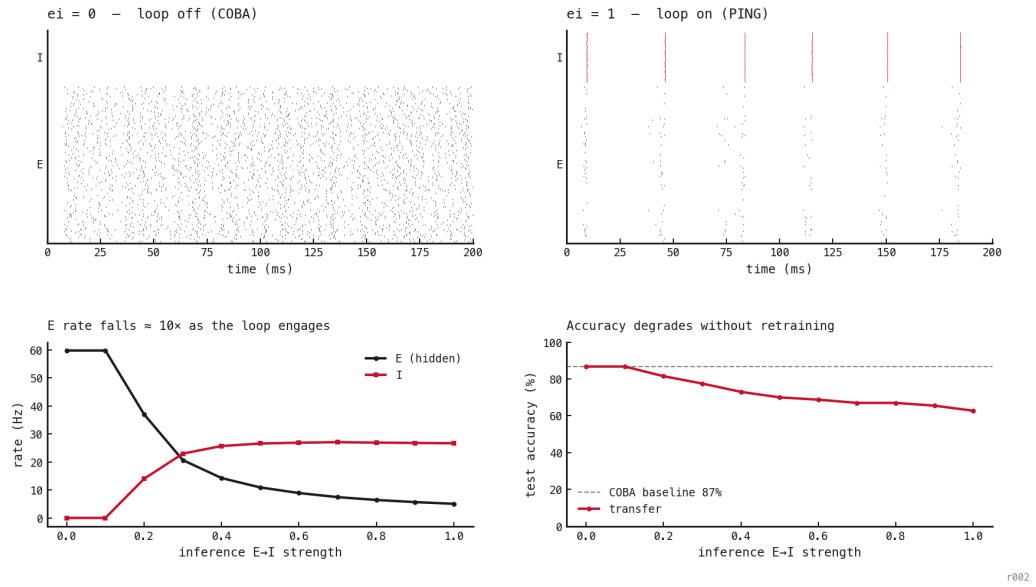


Figure 4:

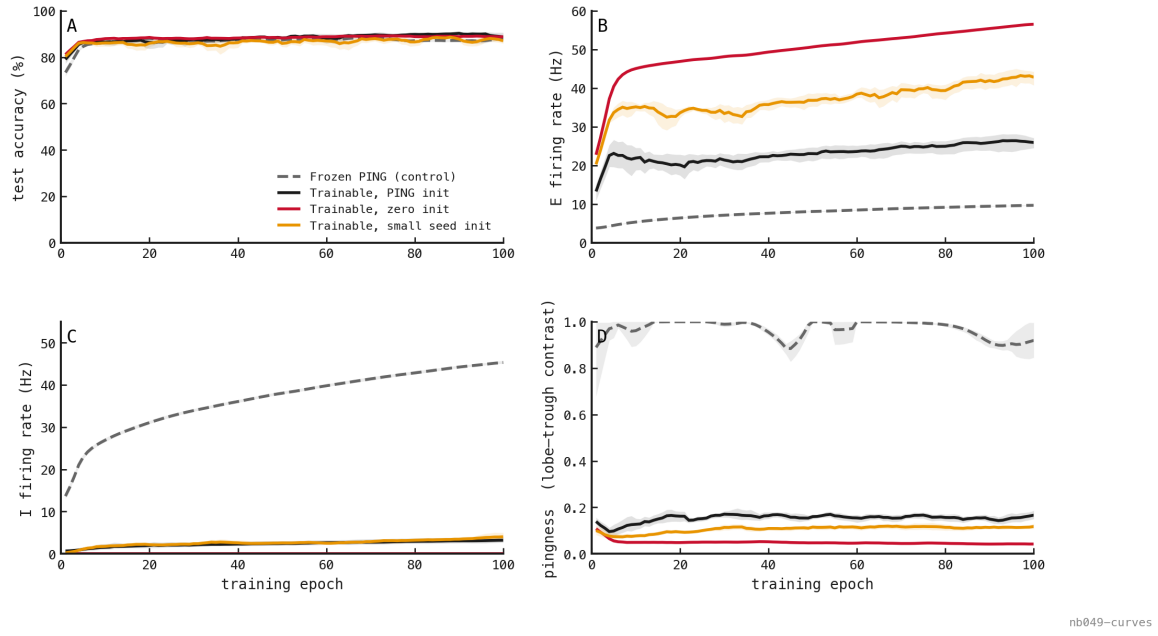


Figure 5:

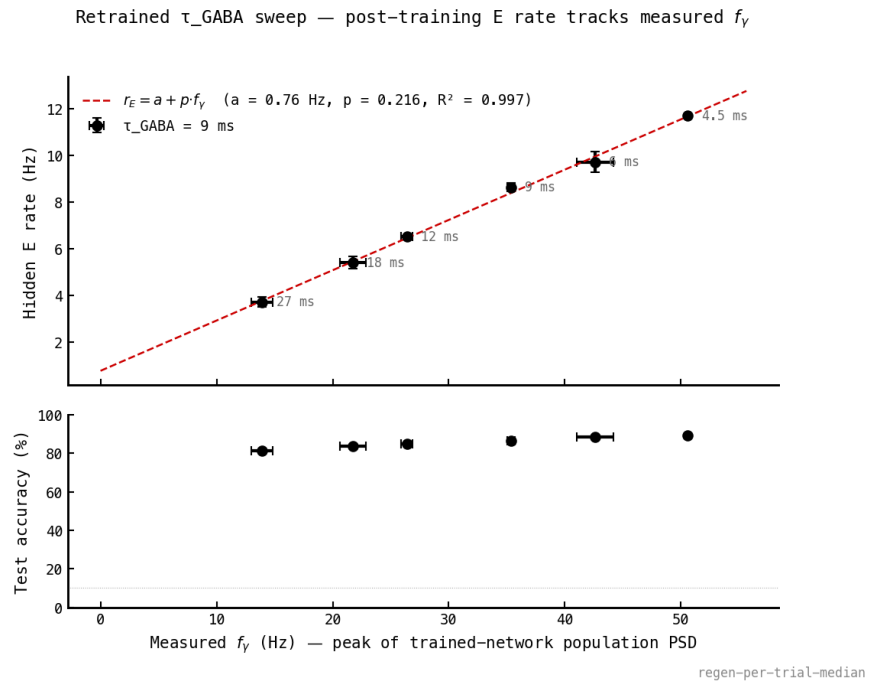


Figure 6:

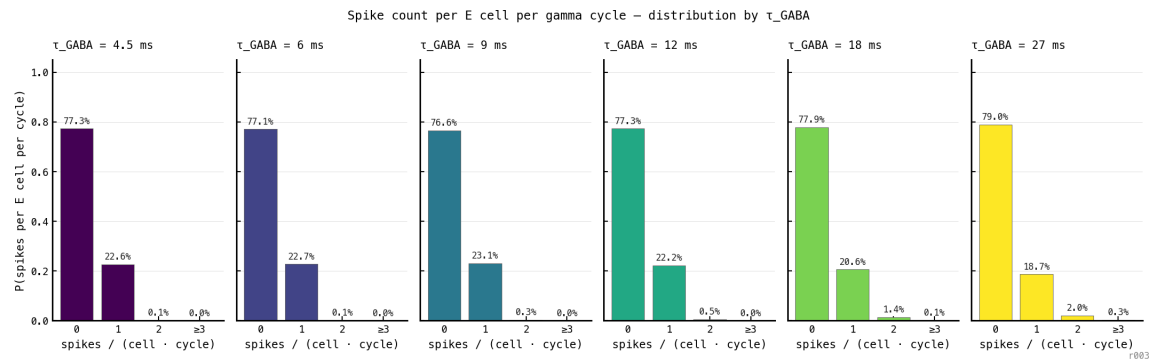


Figure 7:

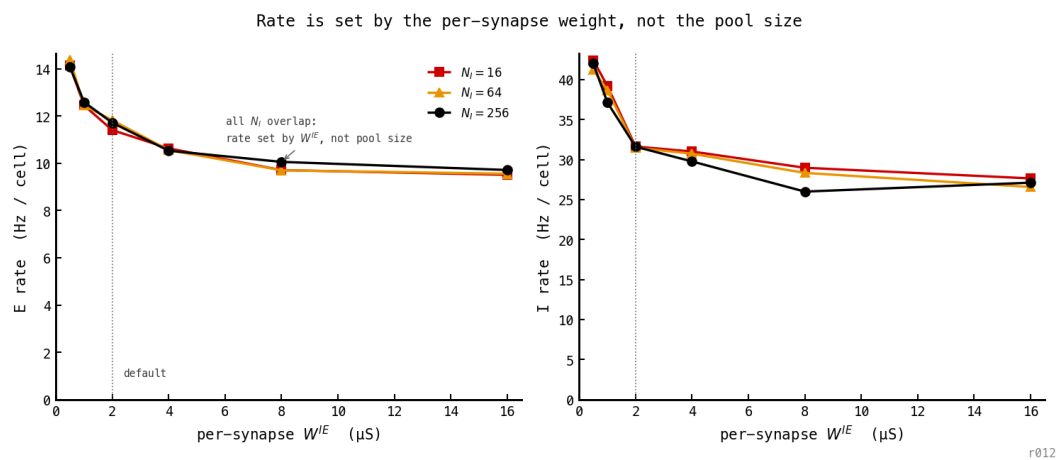


Figure 8:

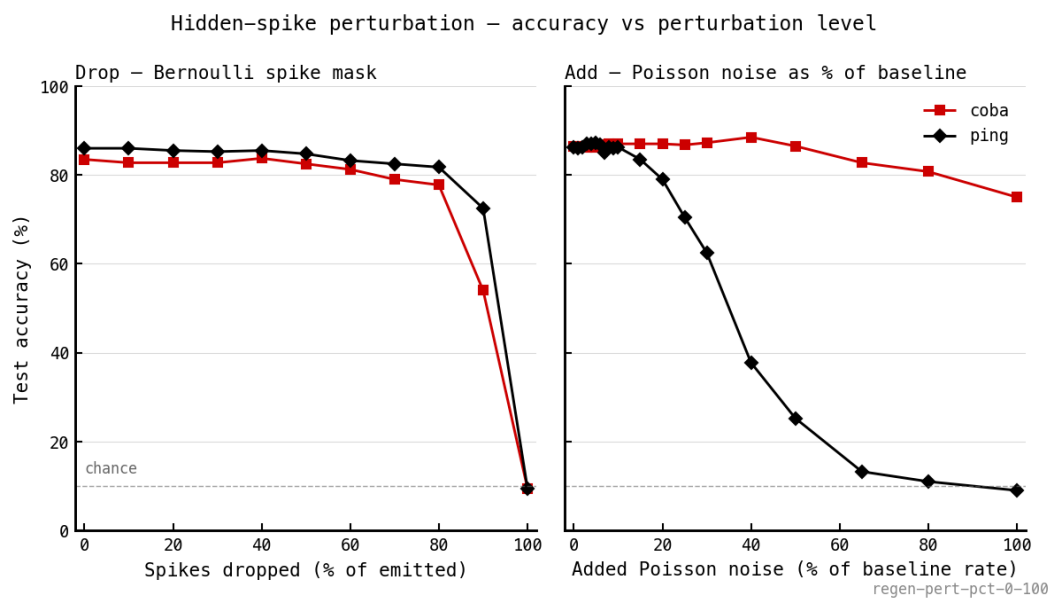


Figure 9:

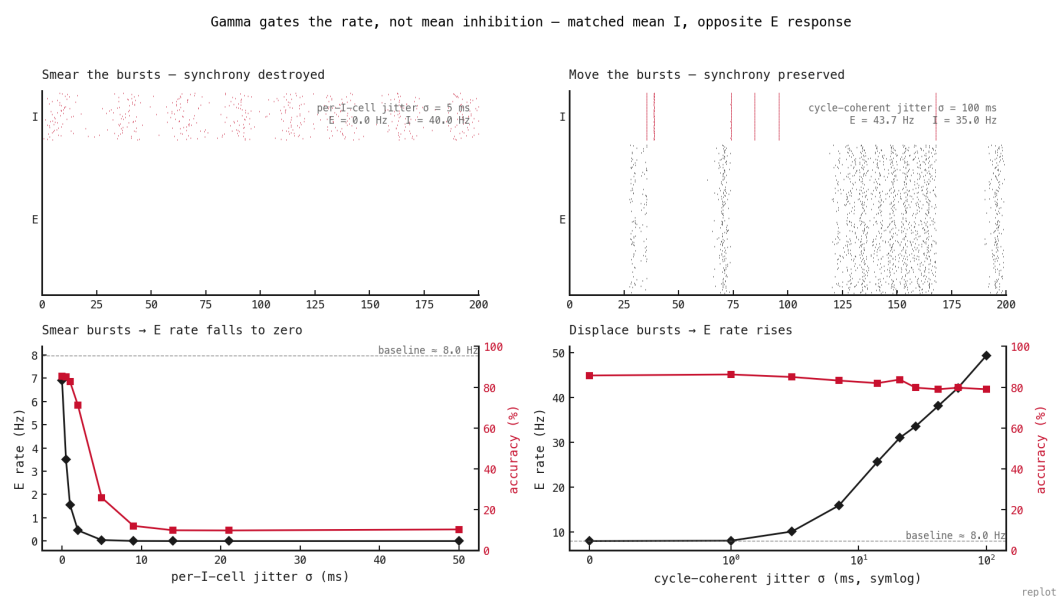


Figure 10:

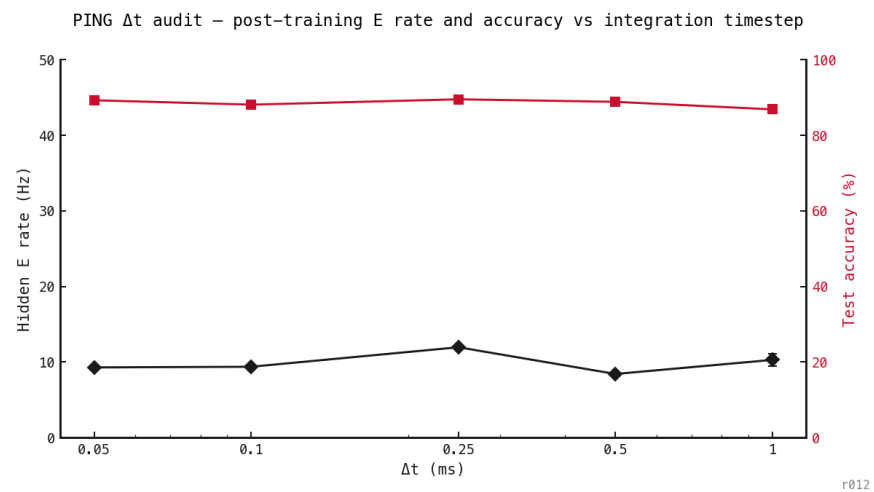


Figure 11:

Streaming digit classification — per-segment (τ , rate) varies, trained PING (no retraining)

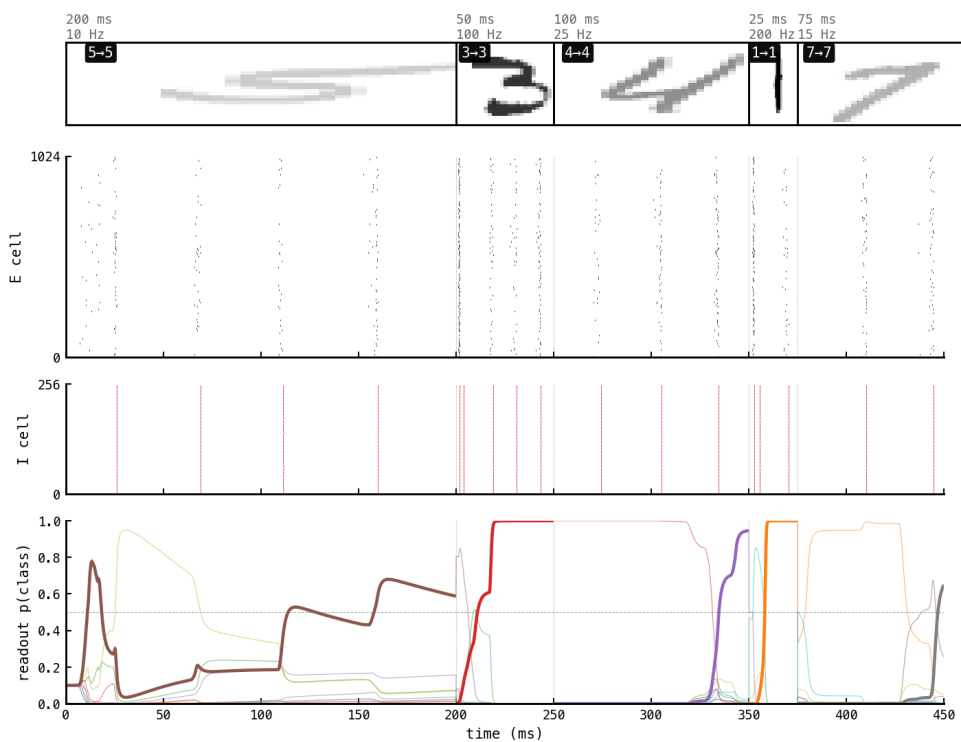


Figure 12:

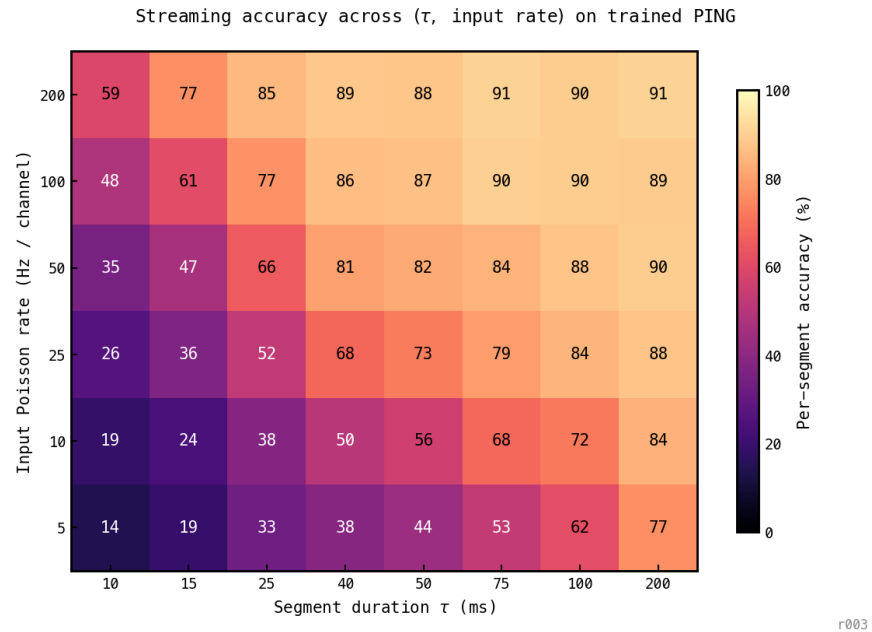


Figure 13: